CSM
Fall 2026

EE 64
Worksheet 8

1. Bode Plots for Filters

Meta: Prereqs: Transfer Functions, Complex magnitude and phases, plotting functions

Bode plots provide us with a simple and easy tool to plot these transfer functions by hand. **Always remember that Bode plots are an approximation**; if you want the precisely correct plots, you need to use numerical methods (like solving using MATLAB or IPython).

When we make Bode plots, we plot the frequency and magintude on a logarithmic scale and the angle in either degrees or radians. We use the log scale because it allows us to break up complex transfer functions into its constituent components.

When plotting the transfer function, the most important quantity to look at is its cutoff frequency ω_c. We will take a look the individual Bode plots for low and high pass filters, and then look at how the Bode plot for a bandpass filter is constructed.

Meta: Note: This semester, drawing straight-line approximations to Bode plots are no longer in scope. However, we will motivate the process of drawing straight line approximations by looking at points for which $\omega \approx 0, \omega = \omega_c$, and $\omega >> \omega_c$.

(a) Let's start by taking a look at $H(j\omega) = X(j\omega)Y(j\omega)$, the product of two transfer functions X and Y.

i. How can you graph $|H(j\omega)|$ using the graphs of $|X(j\omega)|$, $|Y(j\omega)|$?

ii. How does this compare when graphing $\log|H(j\omega)|$ using the graphs of $\log|X(j\omega)|$, $\log|Y(j\omega)|$?

iii. What is the relationship between $\angle H(j\omega), \angle X(j\omega)$, and $\angle Y(j\omega)$?

Solution:

i. To get $H(j\omega)$, we would have to multiply the graphs of $|X(j\omega)|$ and $|Y(j\omega)|$.

ii. Since $\log|H(j\omega)| = \log|X(j\omega)| + \log|Y(j\omega)|$, we would have to add the graphs of $\log X(j\omega)$ and $\log Y(j\omega)$ which is much easier to do.

iii. $H(j\omega) = |H(j\omega)|e^{j\angle H(j\omega)} = |X(j\omega)|e^{j\angle X(j\omega)} \cdot |Y(j\omega)|e^{j\angle Y(j\omega)} = |X(j\omega)||Y(j\omega)|e^{(\angle X(j\omega)+\angle Y(j\omega))}$. Therefore, $\angle H(j\omega) = \angle X(j\omega) + \angle Y(j\omega)$.

(b) Let $H(j\omega) = \frac{1}{1+j\frac{\omega}{\omega_c}}$. This transfer function models the effect of a low pass filter, and has cutoff frequency ω_c.

i. Find $|H(j\omega)|$ and $\angle H(j\omega)$.
Solution: $|H(j\omega)| = \frac{1}{\sqrt{1+(\frac{\omega}{\omega_c})^2}}$. and $\angle H(j\omega) = -\tan^{-1}(\frac{\omega}{\omega_c})$

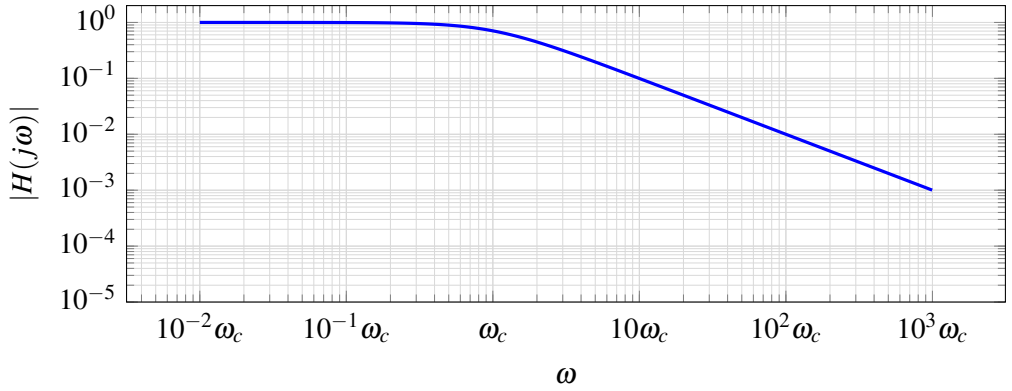
ii. Again, **fill in the following table to analyze the magnitude and phase at the three regions $\omega << \omega_c$, $\omega = \omega_c$, and $\omega >> \omega_c$.**
Solution:

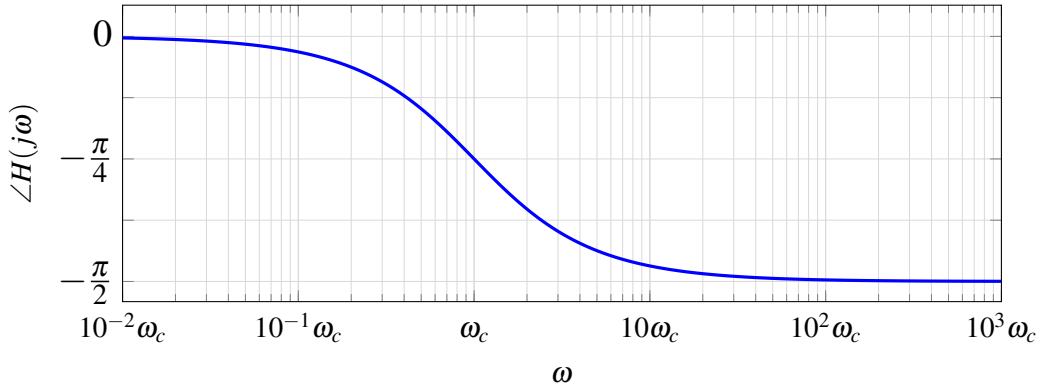
iii. **Draw a sketch of $|H(j\omega)|$ and $\angle H(j\omega)$ using the points, filled in the chart above.** **Solution:**

(c) Let $H(j\omega) = j\omega$. This transfer function does not have a cutoff frequency, but its behavior will prove to be useful when looking at a high-pass filter.

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	$\omega << \omega_c$	$\omega = \omega_c$	$\omega >> \omega_c$
$H_z(\omega)$	1	$\frac{1}{1+j}$	$-j\frac{\omega_c}{\omega}$
$\lvert H_z(\omega)\rvert$	1	$\frac{1}{\sqrt{2}}$	$\frac{\omega_c}{\omega}$
$\angle H_z(\omega)$	0	$-\frac{\pi}{4}$	$-\frac{\pi}{2}$





i. **Find $|H(j\omega)|$ and $\angle H(j\omega)$.**

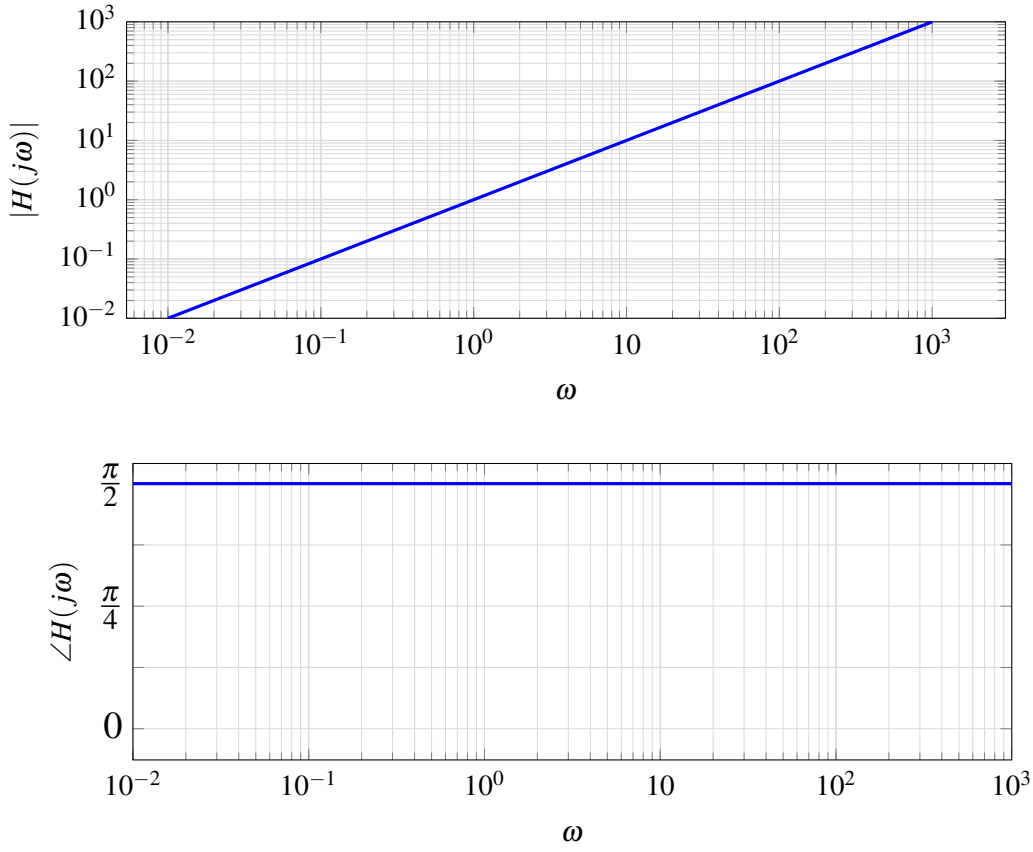
ii. **Sketch them down below.**

Solution:

i. $|H(j\omega)| = \omega$, and $\angle H(j\omega) = \frac{\pi}{2}$.

ii.

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(d) Now let's go back to the low-pass filter with transfer function

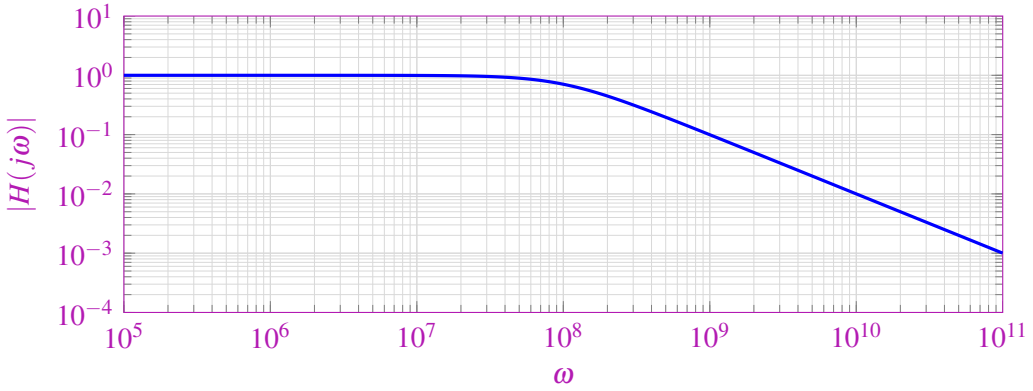
$$H(j\omega) = \frac{1}{j\omega R_1 C_1 + 1} \tag{1}$$

Let $R_1 = 100\Omega$ and $C_1 = 100pF$

i. What is the cutoff frequency of this filter? **Solution:** $\omega_c = \frac{1}{R_1 C_1} = \frac{1}{10^2 \cdot 10^{-10}} = 10^8$

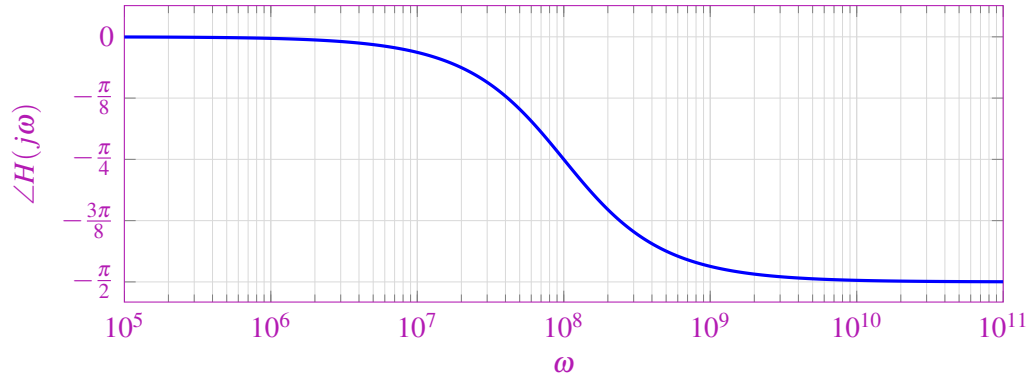
ii. Sketch its phase and magnitude.

Solution: Magnitude (log-log scale): The magnitude of $H(j\omega)$ is close to 1 until ω_c, after which it starts decreasing at a quick rate.



Phase (semi-log scale): The phase of $H(j\omega)$ can be approximated using the three points from the previous parts. When $\omega \approx 0, \angle H(j\omega) \approx 0$, when $\omega = \omega_c, \angle H(j\omega) = \frac{\pi}{4}$, and when $\omega >> \omega_c, \angle H(j\omega) = \frac{\pi}{2}$.

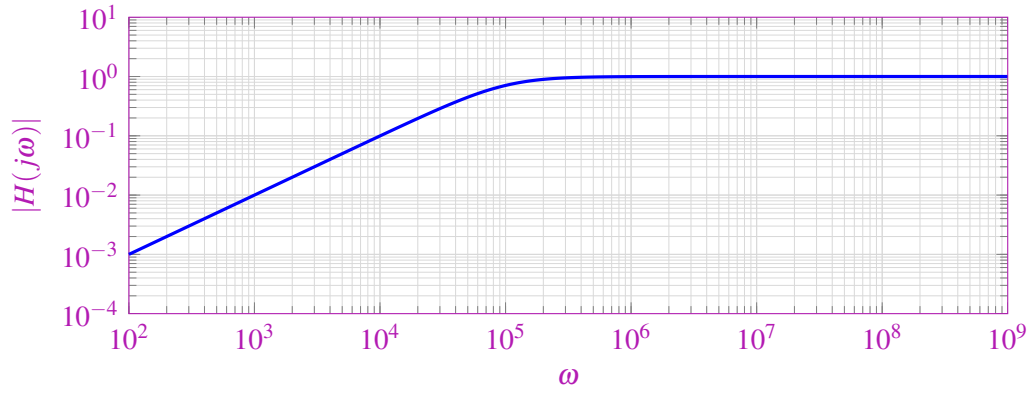
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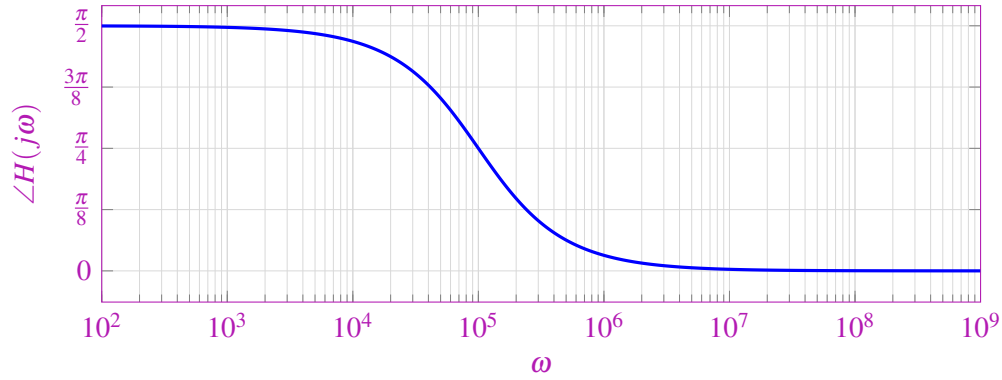
(e) Now consider the transfer function of a High-Pass Filter: $H(j\omega) = \frac{j\omega R_2 C_2}{1+j\omega R_2 C_2}$ where $R_2 = 1k\Omega$ and $C_1 = 10nF$

i. What is the cutoff frequency of this filter? **Solution:** $\omega_c = \frac{1}{R_2 C_2} = \frac{1}{10^3 \cdot 10^{-8}} = 10^5$

ii. Sketch its phase and magnitude.
Solution: Magnitude (log-log scale): The magnitude of $H(j\omega)$ is close to 0 until ω_c, after which it starts increasing at a quick rate, but will asymptotically reach 1 as $\omega >> \omega_c$.



Phase (semi-log scale): The phase of $H(j\omega)$ is $\frac{\pi}{2}$ for $\omega \approx 0$. When $\omega = \omega_c, \angle H(j\omega) = \frac{\pi}{4}$, for larger values of ω, it will continue to decreases until it asymptotically reaches 0 as $\omega >> \omega_c$.



(f) What happens if we cascade these two filters together with a unity-gain buffer in between them? Consider the resulting transfer function: $H(\omega) = \frac{1}{j\omega C_1 R_1 + 1} \cdot \frac{j\omega C_2 R_2}{j\omega C_2 R_2 + 1}$

i. What are the cutoff frequencies of this filter? **Solution:** The lower cutoff is the cutoff frequency of the high-pass filter:

$$\omega_{c,l} = \frac{1}{R_2 C_2} = 10^5$$

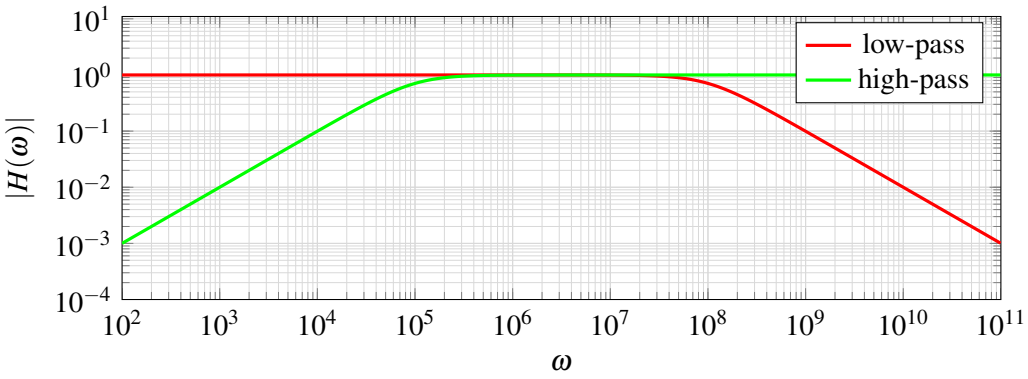
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The upper cutoff is the cutoff frequency of the low-pass filter:

$$\omega_{c,u} = \frac{1}{R_1 C_1} = 10^8$$

ii. Sketch its phase and magnitude. *Hint: How can we combine the plots of the individual filters together?*
Solution: For both the magnitude and the phase plots, you can "add" the plots of the high-pass filter and the low-pass filter. This can be done since both are on a log-log scale.

Magnitude plot:

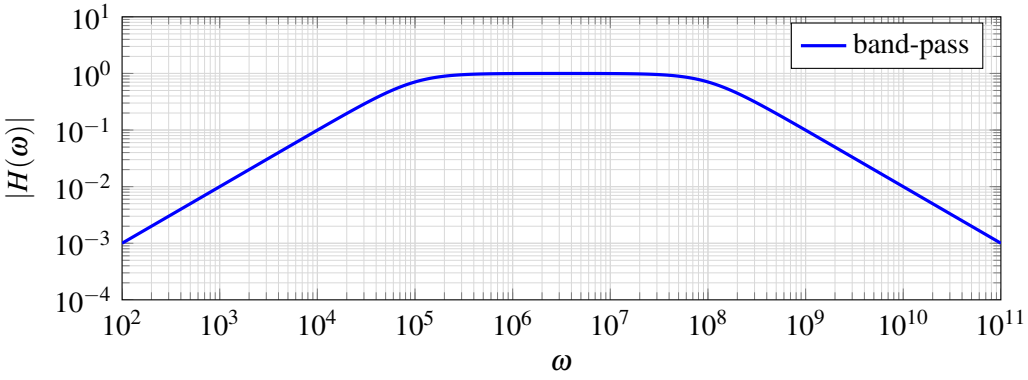


Meta: We can "add" the magnitudes of the high-pass and low-pass filters together because we are graphing $log(|H(\omega)|) = log(|H_{high}(\omega)| \cdot |H_{low}(\omega)|) = log(|H_{high}(\omega)|) + log(|H_{low}(\omega)|)$.
We can "add" the phases because $\angle(H_{high}(\omega) * H_{low}(\omega)) = \angle(H_{high}(\omega)) + \angle(H_{low}(\omega))$.

Now lets take a look at two more concepts, **bandwidth** and **Q-factor**. The bandwidth of a resonance circuit is the difference between its upper and lower cutoff frequencies. It can also be expressed as *resonance frequency* divided by its *Q-factor*:

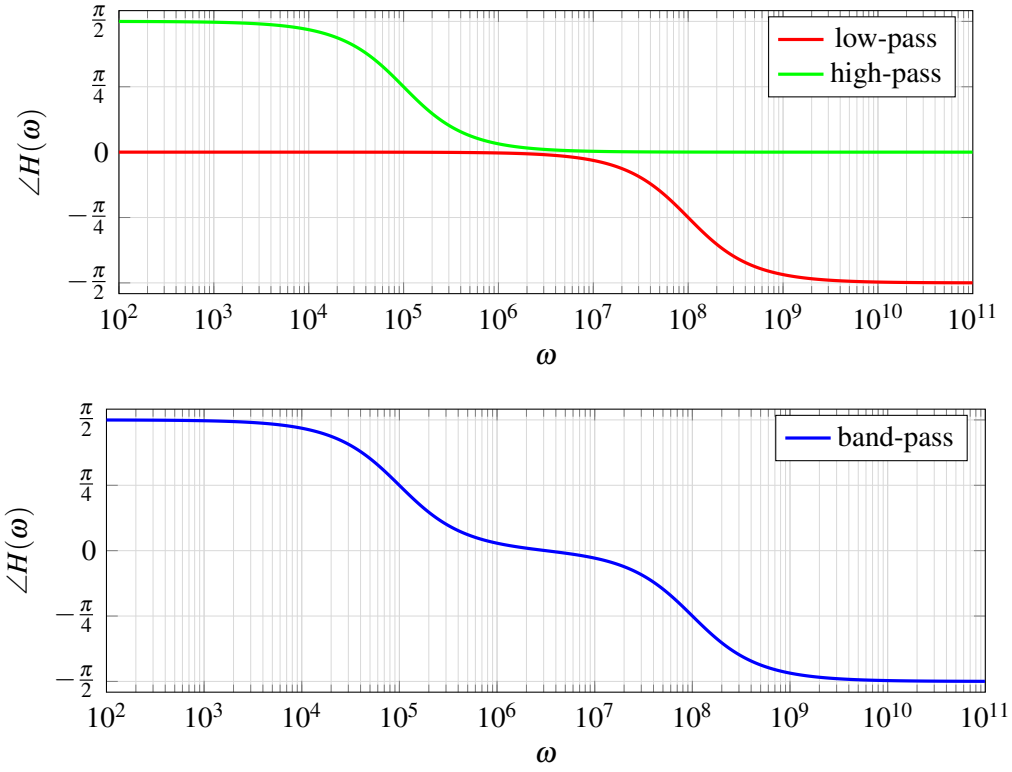
$$bandwidth = \Delta\omega = f_{high} - f_{low} = \frac{\omega_n}{Q} \tag{2}$$

The Q-factor, or quality factor, of a circuit is the ratio of power stored to power dissipated, but can also be more simply thought of as the measure of how good a circuit is, where a higher Q-factor is often more desirable. This can be expressed mathematically as:



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Phase plot:



$$Q = \frac{\omega_n}{\Delta\omega} \tag{3}$$

Meta: Circuits that have low bandwidths relative to their center frequency will have higher Q-factors.

(g) Find the bandwidth of this circuit. **Solution:**

$$\Delta\omega = \omega_{c,u} - \omega_{c,l} = \frac{1}{R_1C_1} - \frac{1}{R_2C_2}$$

$$\Delta\omega = 10^8 - 10^5 = 9.99 \cdot 10^7$$

(h) What is its Q-factor? I.e.: $\frac{\omega_n}{\Delta\omega}$, where ω_n is the center frequency of the band. **Solution:** Symbolically:

$$Q = \frac{\omega_n}{\Delta\omega} = \frac{\frac{1}{2}\cdot\left(\frac{1}{R_1C_1} + \frac{1}{R_2C_2}\right)}{\frac{1}{R_1C_1} - \frac{1}{R_2C_2}}$$

$$Q = \frac{(R_2C_2 + R_1C_1)}{2\cdot(R_2C_2 - R_1C_1)}$$

Numerically:

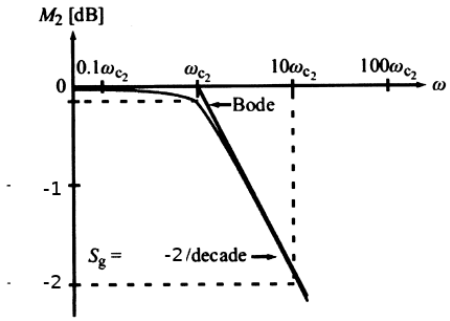
$$Q = \frac{\omega_n}{\Delta\omega} = \frac{1/2(10^8 + 10^5)}{9.99\cdot 10^7} = \frac{5.005\cdot 10^7}{9.99\cdot 10^7}$$

$$Q \approx 0.501$$

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2. Transfer Functions from Bode Plots

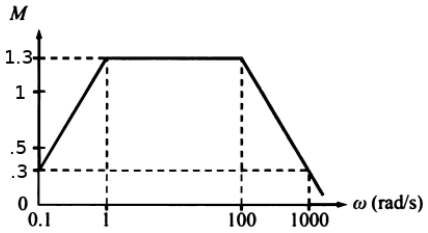
(a) **Derive a transfer function that would result in the following Bode plot.**



Solution:

$$H(\omega) = \frac{1}{\left(1 + \frac{j\omega}{\omega_c}\right)^2}$$

(b) **Derive a transfer function that would result in the following Bode plot.**



Hint: $\log_{10} 2 = 0.3$

Solution:

$$H(\omega) = \frac{j20\omega}{(1 + j\omega)\left(1 + \frac{j\omega}{100}\right)}$$

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3. Op Amps as Buffers

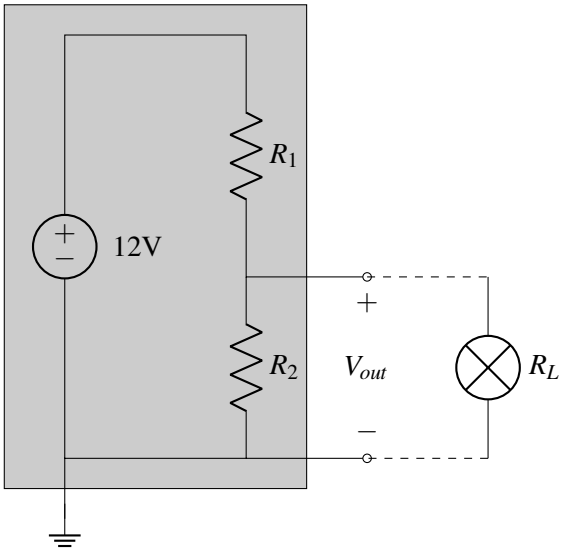
Learning Goal: This problem helps understand the operating principle of an op-amp buffer and how it helps with loading.

Relevant Notes: *Note 19 Section 19.7* goes different op-amp circuit topology and corresponding derivations.

Meta: Show that the loading effect becomes less pronounced when the load resistance is higher, i.e. the load is drawing less current.

Now we will revisit a problem that you might have seen before, with our new knowledge of op-amps. We have access to a circuit inside a 'black box' as shown below, with two terminal coming out of it.

(a) We need a voltage of 6V power a light bulb with resistance R_L. Design R_1 and R_2 inside the black box so that the voltage across R_2 is exactly equal to this required voltage **when the bulb is not connected**; i.e. $V_{R_2} = V_{out} = 6\text{V}$.



Solution: The voltage across R_2 is given by

$$V_{R_2} = \frac{R_2}{R_1 + R_2} \times 12\text{V}$$

If we set $V_{R_2} = 6\text{V}$, we get

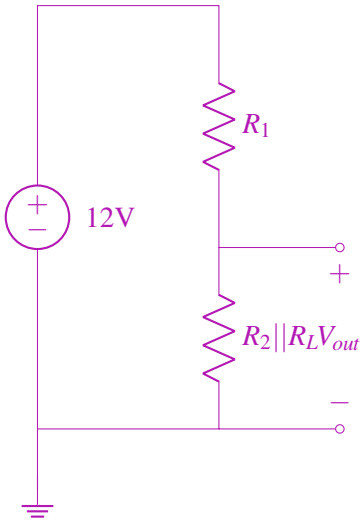
$$6\text{V} = \frac{R_2}{R_1 + R_2} \times 12\text{V}$$
$$\implies R_1 = R_2$$

For example, we can choose $R_1 = R_2 = 1\text{k}\Omega$.

(b) Now let us connect the bulb R_L across R_2. What is the voltage across R_1, R_2 and R_L when the bulb is connected when $R_L = R_2$? Use the values of R_1 and R_2 from the last part. Will the light bulb turn on? What happens if $R_L = 2R_2$?

Solution: When we connect the bulb across R_2, the equivalent circuit is the following:

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The voltage V_{out} given by

$$V_{out} = \frac{R_2||R_L}{R_1 + R_2||R_L} \times 12\text{V}$$

When $R_L = R_2$:

$$R_2||R_L = R_2||R_2 = \frac{R_2}{2}.$$

Hence V_{out} is:

$$V_{out} = \frac{R_2||R_L}{R_1 + R_2||R_L} \times 12\text{V} = \frac{\frac{R_2}{2}}{R_1 + \frac{R_2}{2}} \times 12\text{V} = \frac{\frac{R_2}{2}}{R_2 + \frac{R_2}{2}} \times 12\text{V} = 4\text{V} < 6\text{V}$$

So the bulb will not turn on.

When $R_L = 2R_2$:

$$R_2||R_L = R_2||2R_2 = \frac{2R_2}{3}.$$

Hence V_{out} is:

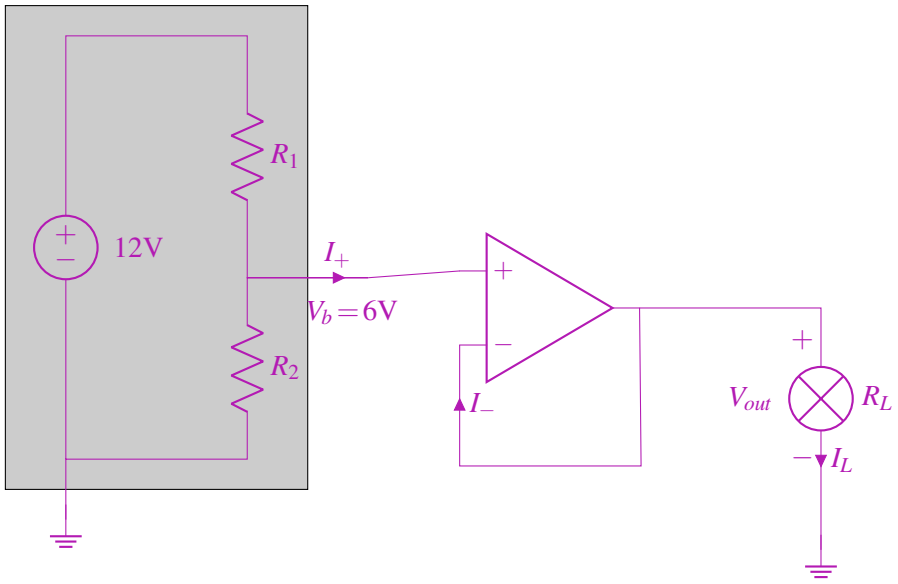
$$V_{out} = \frac{R_2||R_L}{R_1 + R_2||R_L} \times 12\text{V} = \frac{\frac{2R_2}{3}}{R_1 + \frac{2R_2}{3}} \times 12\text{V} = \frac{\frac{2R_2}{3}}{R_2 + \frac{2R_2}{3}} \times 12\text{V} = 4.8\text{V} < 6\text{V}$$

Although, V_{out} is now closer to 6V, the bulb will still not turn on.

(c) Using your knowledge of op-amps, how could you resolve this issue of V_{out} changing based on the value of R_L? Think about how you might use an op-amp buffer.

Solution: We can introduce a 'buffer' op-amp as shown below:

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A buffer op amp effectively 'decouples' its input and output, It does this by preventing the bulb from drawing current from the black box circuit. Remember that the input currents of an ideal op-amp:

$$I_+ = I_- = 0,$$

while the output current I_L can adjust itself to any extent, depending on the demand placed by V_{out}. Let us use nodal analysis and the golden rules to formally solve this circuit. Firstly, we observe that the op-amp is in negative feedback configuration. Using the golden rules, we have that $V_+ = V_- = V_b = 6V$. Also, because the feedback connection is a short, $V_{out} = V_-$. Therefore, $V_{out} = V_b = 6V$. This is exactly what we want! The voltage across R_L equals $V_{out} = 6V$, which is above the required voltage.

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